UNIVERSITY OF CAMBRIDGE INTERNATIONAL EXAMINATIONS
General Certificate of Education Ordinary Level

CANDIDATE NAME

CENTRE NUMBER

CANDIDATE NUMBER

* 3 4 6 8 3 4 0 1 1 7 *

CHEMISTRY **5070/04**

Paper 4 Alternative to Practical **May/June 2008**

1 hour

Candidates answer on the Question Paper.

No additional materials are required.

READ THESE INSTRUCTIONS FIRST

Write your Centre number, candidate number and name on all the work you hand in.
Write in dark blue or black pen.
You may use a soft pencil for any diagrams, graphs or rough working.
Do **not** use staples, paper clips, highlighters, glue or correction fluid.
DO **NOT** WRITE IN ANY BARCODES.

Answer **all** questions.
Write your answers in the spaces provided in the Question Paper.
The number of marks is given in brackets [ ] at the end of each question or part question.
At the end of the examination, fasten all your work securely together.

For Examiner's Use

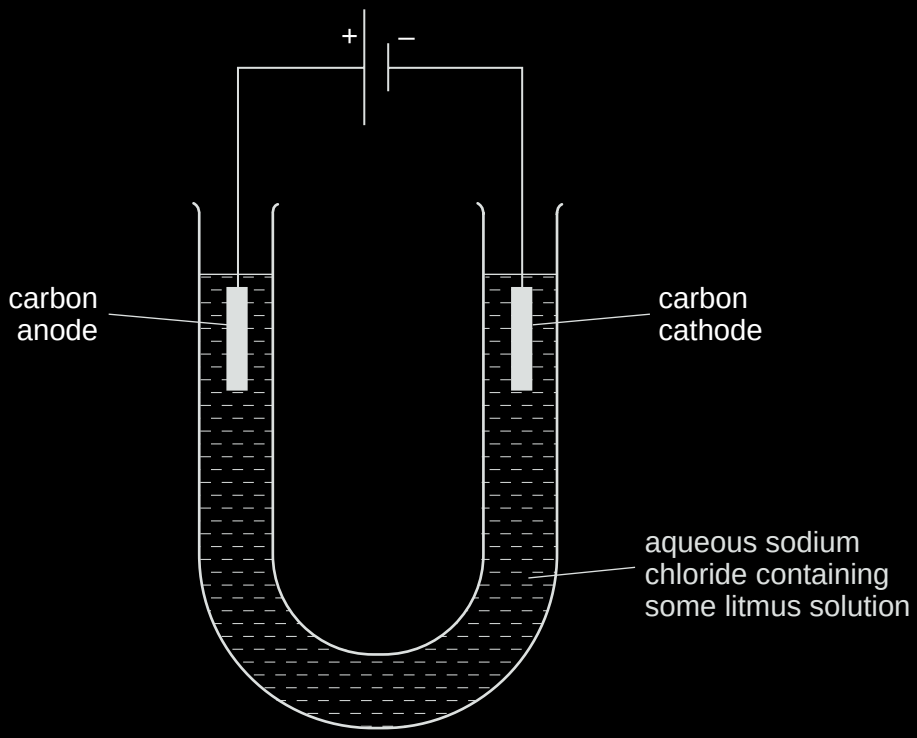
This document consists of **18** printed pages and **2** blank pages.

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.....

[Total: 1]



(a) (i) Name the gas produced at the anode (positive electrode).

.....[1]

(ii) Suggest what happened to the colour of the solution around the anode as the electrolysis proceeded.

.....[1]

(iii) Why did this change take place?

.....[1]

(b) (i) Name the gas produced at the cathode (negative electrode).

.....[1]

(ii) Give a test for this gas.

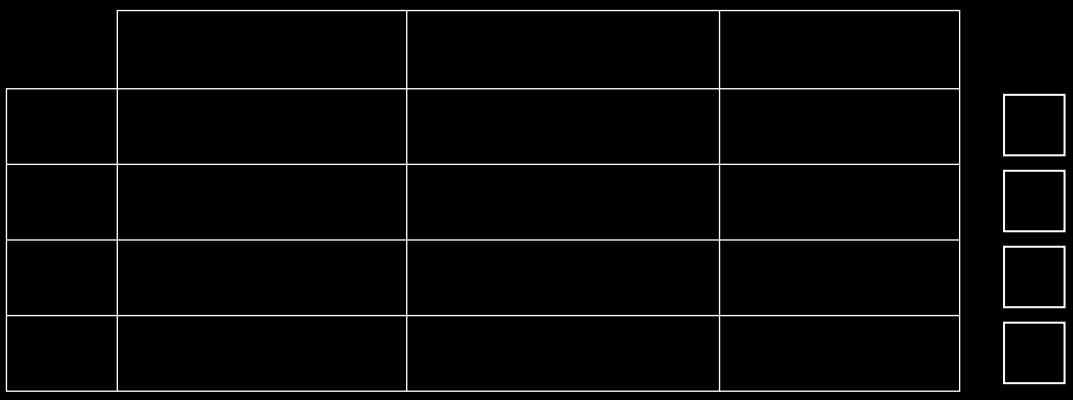
.....[1]

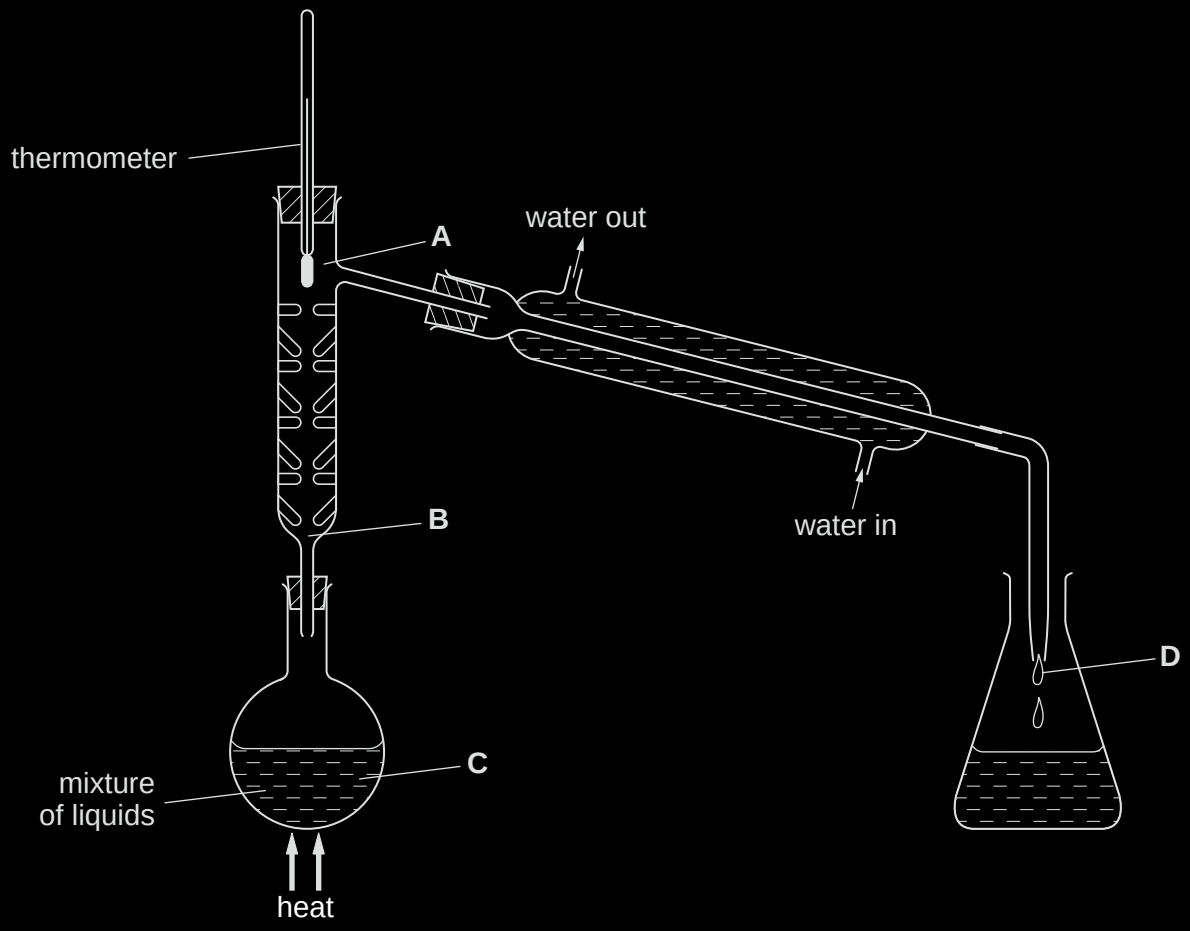
(iii) What happened to the colour of the solution around the cathode as the electrolysis proceeded?

.....[1]

(iv) Why did this change take place?

.....[1]





When the thermometer first shows a steady reading, at which point **A**, **B**, **C**, or **D** will there be the highest proportion of the liquid with the higher boiling point?

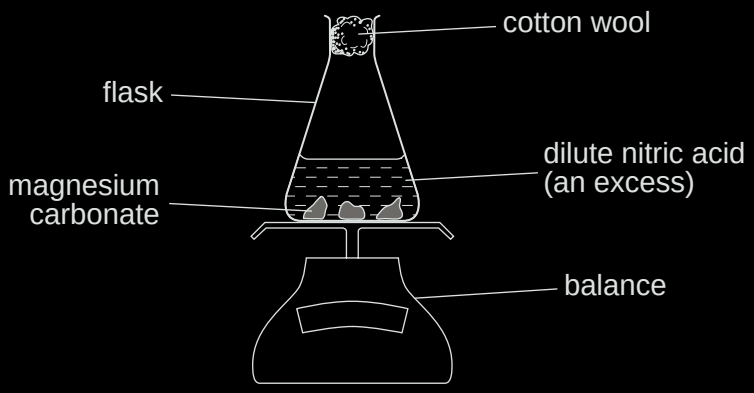
(a) **A** ☐

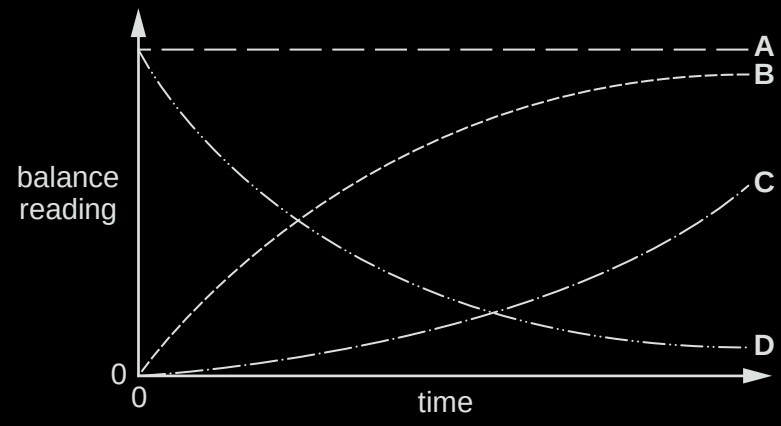
(b) **B** ☐

(c) **C** ☐

(d) **D** ☐

[Total: 1]





A graph of the balance reading against time was plotted.
Which curve was obtained?

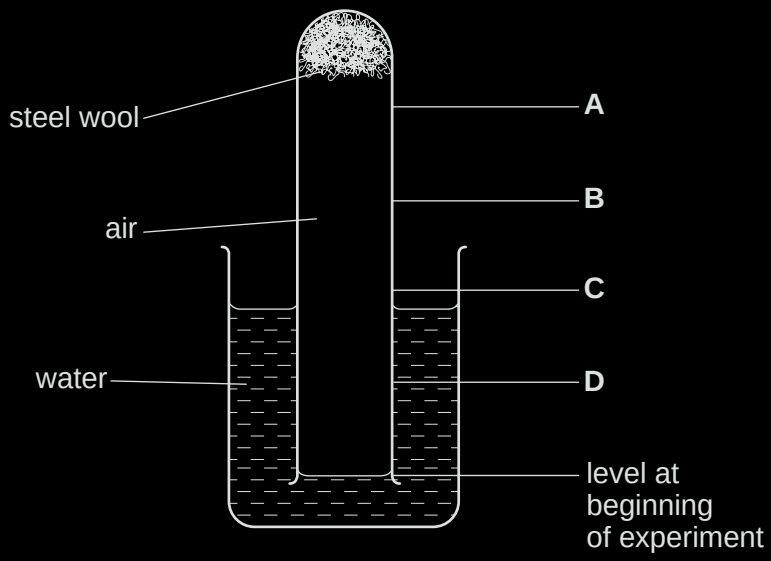
(a) **A** ☐

(b) **B** ☐

(c) **C** ☐

(d) **D** ☐

[Total: 1]



What was the level of the water after several days?

(a) **A** ☐

(b) **B** ☐

(c) **C** ☐

(d) **D** ☐

[Total: 1]

8 Calcium carbonate reacts with hydrochloric acid as shown below.

$$CaCO_3 + 2HCl \rightarrow CaCl_2 + CO_2 + H_2O$$

What volume of 1.0 mol/dm^3 hydrochloric acid is needed to completely react with 2.0 g of calcium carbonate?
[M_r: $CaCO_3$, 100]

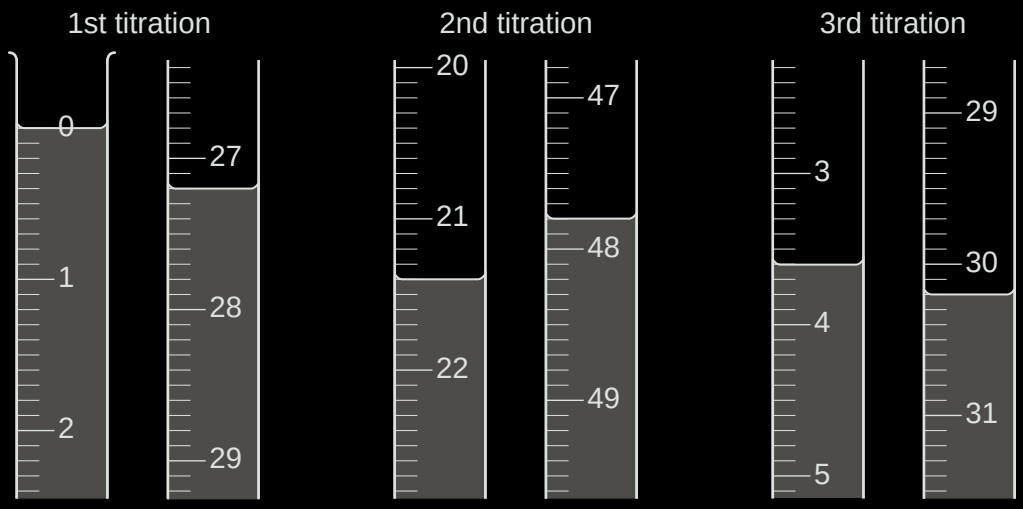
(a) 20 cm^3 ☐

(b) 40 cm^3 ☐

(c) 200 cm^3 ☐

(d) 400 cm^3 ☐

[Total: 1]



(d) Use the diagrams to complete the following table.

titration number	1	2	3
final burette reading /cm^3			
initial burette reading /cm^3			
volume of **Q** used /cm^3			
best titration results (✓)			

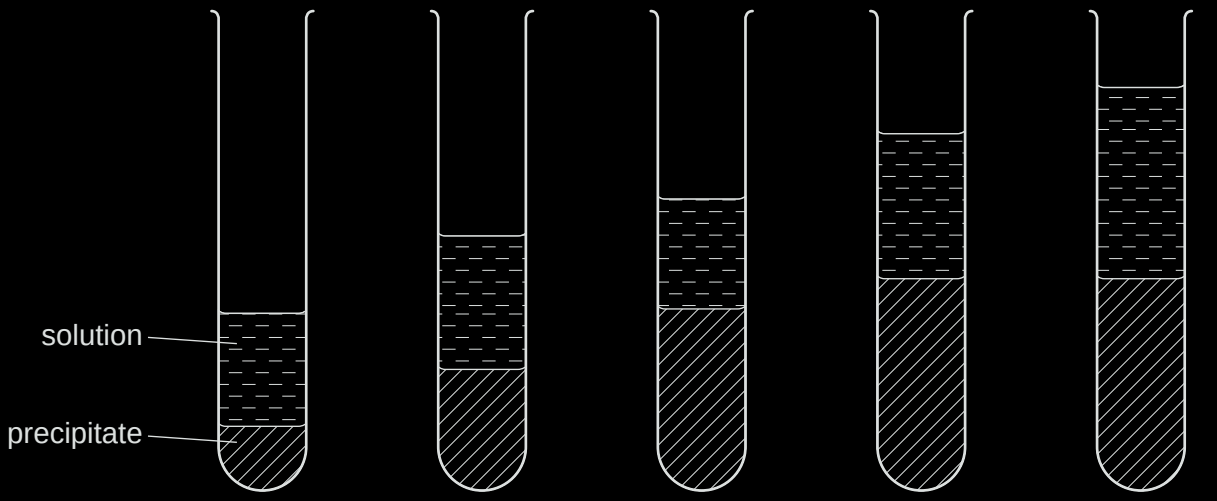
Summary:

Tick (✓) the best titration results.

Using these results the average volume of **Q** was .....cm^3. [4]

(e) **Q** was 0.0800 mol/dm^3 hydrochloric acid. Calculate the number of moles of hydrochloric acid in the average volume of **Q** calculated in **(d)**.

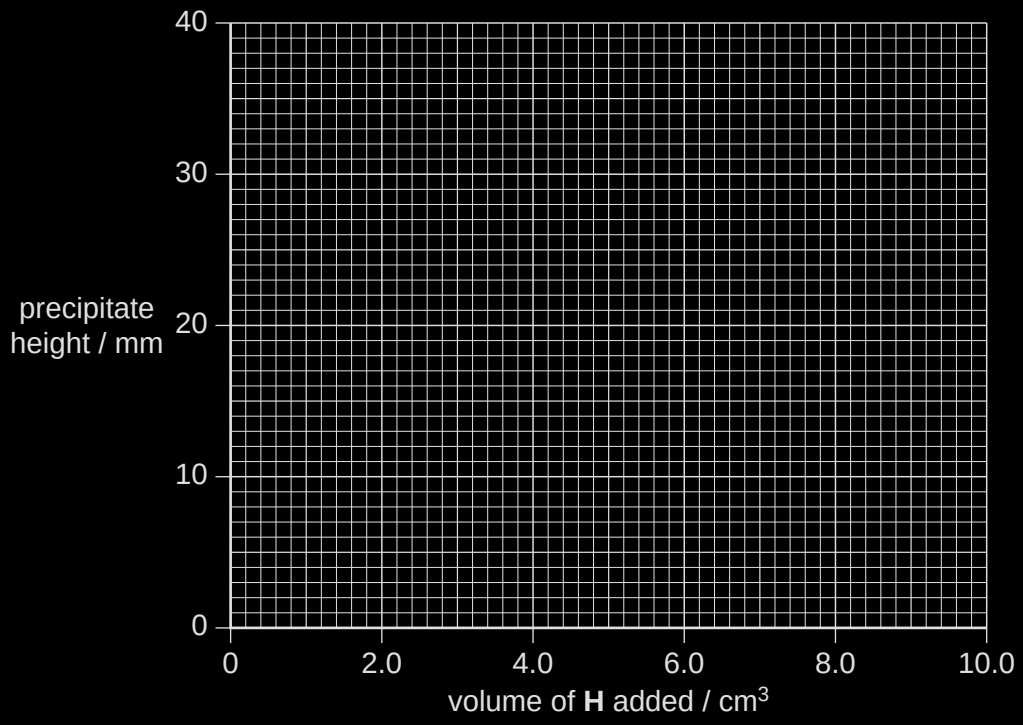
..... [1]



(b) Measure the height of each precipitate in millimetres. Record the results in the table below.

test-tube	1	2	3	4	5
volume of **H** added / cm^3	2.0	4.0	6.0	8.0	10.0
precipitate height / mm					

[2]



[3]

(d) **(i)** What was the minimum volume of **H** required to completely react with 4.0 cm^3 **G**?

.....cm^3 [1]

(ii) Using your answers to **(a)** and **(d)(i)**, calculate the concentration of potassium iodide in **G**.

.....mol/dm^3 [2]